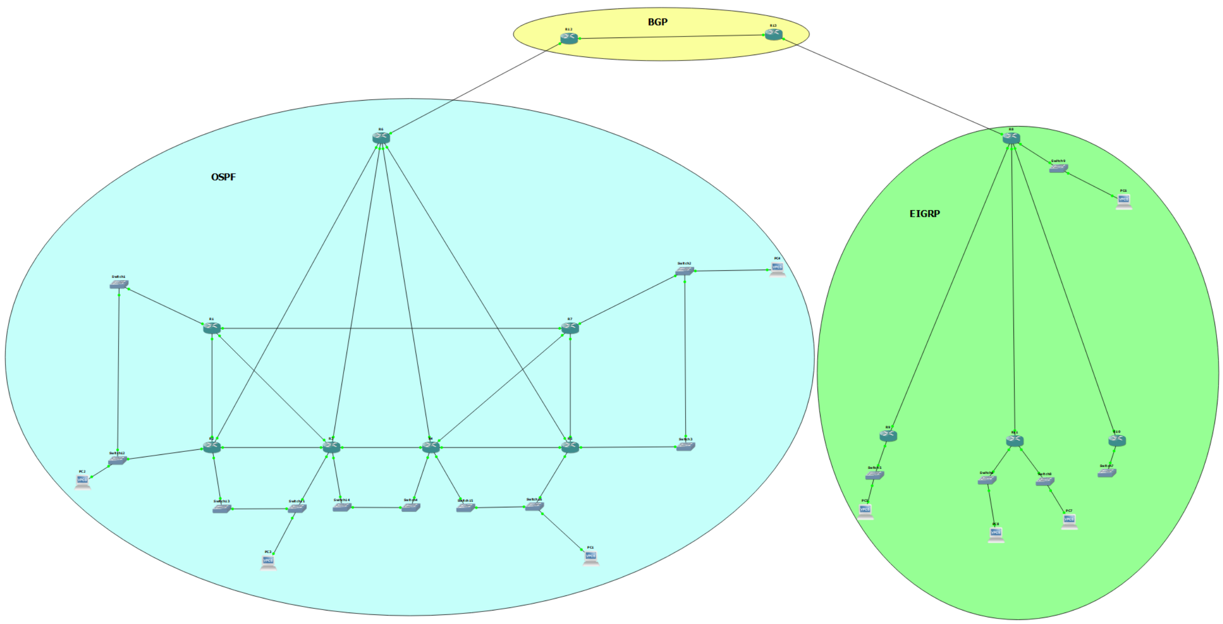
PART 1



For this Lab, I had chosen to do VRRP, or virtual router redundancy protocol, and BGP, or border gateway protocol. I had my two previous topologies made in lab 6 and just reconfigured and changed them a bit.

The first topology which is Topology B is configured with OSPF routing and the second topology which is A is configured with EIGRP.

The image you are seeing above is the final topology except for the VPCs which I will add more later but for testing purposes, these were enough

Virtual Router Redundancy Protocol (VRRP)

This was one of the configurations I had made for VRRP in one of the routers with a virtual address of 192.168.1.4. And I had made these in two routers including this one.

This acts as a master router and the other as a backup with the same group number. When this shuts down the backup will take the master position and continue routing traffic between the networks or within and hence no subnet is taken offline.

R1

```
R1#sh vrrp
FastEthernet0/0 - Group 1
  State is Master
  Virtual IP address is 192.168.1.4
  Virtual MAC address is 0000.5e00.0101
  Advertisement interval is 1.000 sec
  Preemption enabled
  Priority is 100
  Master Router is 192.168.1.6 (local), priority is 100
  Master Advertisement interval is 1.000 sec
  Master Down interval is 3.609 sec
```

R2

```
R2#sh vrrp
FastEthernet0/0 - Group 2
  State is Master
  Virtual IP address is 192.168.1.12
  Virtual MAC address is 0000.5e00.0102
  Advertisement interval is 1.000 sec
  Preemption enabled
  Priority is 100
  Master Router is 192.168.1.14 (local), priority is 100
  Master Advertisement interval is 1.000 sec
  Master Down interval is 3.609 sec

FastEthernet0/1 - Group 1
  State is Backup
  Virtual IP address is 192.168.1.4
  Virtual MAC address is 0000.5e00.0101
  Advertisement interval is 1.000 sec
  Preemption enabled
  Priority is 100
  Master Router is 192.168.1.6, priority is 100
  Master Advertisement interval is 1.000 sec
  Master Down interval is 3.609 sec (expires in 3.273 sec)
```

R3

```
R3#sh vrrp
FastEthernet0/0 - Group 2
  State is Backup
  Virtual IP address is 192.168.1.12
  Virtual MAC address is 0000.5e00.0102
  Advertisement interval is 1.000 sec
  Preemption enabled
  Priority is 100
  Master Router is 192.168.1.14, priority is 100
  Master Advertisement interval is 1.000 sec
  Master Down interval is 3.609 sec (expires in 2.777 sec)

FastEthernet0/1 - Group 3
  State is Master
  Virtual IP address is 192.168.1.20
  Virtual MAC address is 0000.5e00.0103
  Advertisement interval is 1.000 sec
  Preemption enabled
  Priority is 100
  Master Router is 192.168.1.22 (local), priority is 100
  Master Advertisement interval is 1.000 sec
  Master Down interval is 3.609 sec
```

R4

```
R4#sh vrrp
FastEthernet0/0 - Group 3
  State is Backup
  Virtual IP address is 192.168.1.20
  Virtual MAC address is 0000.5e00.0103
  Advertisement interval is 1.000 sec
  Preemption enabled
  Priority is 100
  Master Router is 192.168.1.22, priority is 100
  Master Advertisement interval is 1.000 sec
  Master Down interval is 3.609 sec (expires in 2.617 sec)

FastEthernet0/1 - Group 4
  State is Master
  Virtual IP address is 192.168.1.28
  Virtual MAC address is 0000.5e00.0104
  Advertisement interval is 1.000 sec
  Preemption enabled
  Priority is 100
  Master Router is 192.168.1.30 (local), priority is 100
  Master Advertisement interval is 1.000 sec
  Master Down interval is 3.609 sec
```

R5

```
R5#sh vrrp
FastEthernet0/0 - Group 5
  State is Master
  Virtual IP address is 192.168.1.36
  Virtual MAC address is 0000.5e00.0105
  Advertisement interval is 1.000 sec
  Preemption enabled
  Priority is 100
  Master Router is 192.168.1.38 (local), priority is 100
  Master Advertisement interval is 1.000 sec
  Master Down interval is 3.609 sec

FastEthernet0/1 - Group 4
  State is Backup
  Virtual IP address is 192.168.1.28
  Virtual MAC address is 0000.5e00.0104
  Advertisement interval is 1.000 sec
  Preemption enabled
  Priority is 100
  Master Router is 192.168.1.30, priority is 100
  Master Advertisement interval is 1.000 sec
  Master Down interval is 3.609 sec (expires in 2.649 sec)
```

Configurations

As I previously mentioned I had configured OSPF for topology B and this is just one of the router neighbors that it is detecting.

```
R1#sh ip ospf neigh

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.1.205     1   FULL/BDR        00:00:37    192.168.1.205   FastEthernet2/0
192.168.1.197     1   FULL/BDR        00:00:35    192.168.1.197   FastEthernet0/1
192.168.1.189     1   FULL/BDR        00:00:34    192.168.1.189   FastEthernet1/0
192.168.1.197     1   FULL/BDR        00:00:36    192.168.1.5     FastEthernet0/0
```

This one is for Topology A which I used EIGRP and it is able to detect the routers nearby fine.

```
R4#sh ip eigrp neigh
IP-EIGRP neighbors for process 10
H   Address                 Interface       Hold Uptime   SRTT   RTO  Q  Seq
                                            (sec)         (ms)       Cnt Num
2   192.168.0.198           Fa0/1              5 00:27:55   66   396  0  11
1   192.168.0.206           Fa0/0              6 00:27:56   70   420  0  13
0   192.168.0.190           Fa1/0              6 00:27:56   73   438  0  14
```

VRRP working

Over here we can see VRRP in action where when pinging to a different network's PC and suddenly a router were to stop we can see that after a few seconds, the pinging kicks back in and the backup router is in effect which means it works!!
The reason it takes a while is because of the OSPF routing which takes time to update the routing table and find the most efficient path to take to reach the destination

```
PC2> ping 192.168.0.1 -t

84 bytes from 192.168.0.1 icmp_seq=1 ttl=56 time=231.223 ms
84 bytes from 192.168.0.1 icmp_seq=2 ttl=56 time=225.671 ms
84 bytes from 192.168.0.1 icmp_seq=3 ttl=56 time=221.571 ms
84 bytes from 192.168.0.1 icmp_seq=4 ttl=56 time=223.161 ms
84 bytes from 192.168.0.1 icmp_seq=5 ttl=56 time=223.094 ms
84 bytes from 192.168.0.1 icmp_seq=6 ttl=56 time=221.682 ms
84 bytes from 192.168.0.1 icmp_seq=7 ttl=56 time=223.046 ms
192.168.0.1 icmp_seq=8 timeout
192.168.0.1 icmp_seq=9 timeout
192.168.0.1 icmp_seq=10 timeout
192.168.0.1 icmp_seq=11 timeout
192.168.0.1 icmp_seq=12 timeout
192.168.0.1 icmp_seq=13 timeout
192.168.0.1 icmp_seq=14 timeout
192.168.0.1 icmp_seq=15 timeout
192.168.0.1 icmp_seq=16 timeout
192.168.0.1 icmp_seq=17 timeout
192.168.0.1 icmp_seq=18 timeout
192.168.0.1 icmp_seq=19 timeout
192.168.0.1 icmp_seq=20 timeout
192.168.0.1 icmp_seq=21 timeout
192.168.0.1 icmp_seq=22 timeout
192.168.0.1 icmp_seq=23 timeout
192.168.0.1 icmp_seq=24 timeout
192.168.0.1 icmp_seq=25 timeout
192.168.0.1 icmp_seq=26 timeout
192.168.0.1 icmp_seq=27 timeout
192.168.0.1 icmp_seq=28 timeout
192.168.0.1 icmp_seq=29 timeout
192.168.0.1 icmp_seq=30 timeout
192.168.0.1 icmp_seq=31 timeout
84 bytes from 192.168.0.1 icmp_seq=32 ttl=56 time=217.059 ms
84 bytes from 192.168.0.1 icmp_seq=33 ttl=56 time=213.611 ms
84 bytes from 192.168.0.1 icmp_seq=34 ttl=56 time=205.285 ms
84 bytes from 192.168.0.1 icmp_seq=35 ttl=56 time=208.780 ms
84 bytes from 192.168.0.1 icmp_seq=36 ttl=56 time=203.735 ms
84 bytes from 192.168.0.1 icmp_seq=37 ttl=56 time=208.818 ms
84 bytes from 192.168.0.1 icmp_seq=38 ttl=56 time=204.894 ms
84 bytes from 192.168.0.1 icmp_seq=39 ttl=56 time=206.470 ms
84 bytes from 192.168.0.1 icmp_seq=40 ttl=56 time=207.127 ms
```

Border Gateway Protocol (BGP)

We can see that R6 in topology B has all the routes available to reach every part of every network. The way we do this is through the **redistribution** of protocols.

Since we are using different routing protocols in two different topologies they will not be able to share the routing information with each other that easily so we redistribute so that other protocols can see the routes discovered by that protocol and hence redistribution.

```
   Network          Next Hop            Metric LocPrf Weight Path
*  10.7.1.0/24      10.7.1.101               0             0 200 ?
*>                  0.0.0.0                  0         32768 ?
*> 20.0.1.0/24      10.7.1.101                             0 200 ?
*> 92.1.16.0/24     10.7.1.101               0             0 200 i
*> 192.168.0.0/29   10.7.1.101                             0 200 300 ?
*> 192.168.0.8/29   10.7.1.101                             0 200 300 ?
*> 192.168.0.16/29  10.7.1.101                             0 200 300 ?
*> 192.168.0.24/29  10.7.1.101                             0 200 300 ?
*> 192.168.0.40/29  10.7.1.101                             0 200 300 ?
*> 192.168.0.184/29 10.7.1.101                             0 200 300 ?
*> 192.168.0.192/29 10.7.1.101                             0 200 300 ?
*> 192.168.0.200/29 10.7.1.101                             0 200 300 ?
*> 192.168.1.0/29   192.168.1.102           60         32768 ?
*> 192.168.1.8/29   192.168.1.110           60         32768 ?
*> 192.168.1.16/29  192.168.1.110           60         32768 ?
*> 192.168.1.24/29  192.168.1.102           60         32768 ?
*> 192.168.1.32/29  192.168.1.102           60         32768 ?
*> 192.168.1.96/29  0.0.0.0                  0         32768 i
*> 192.168.1.104/29 0.0.0.0                  0         32768 i
*> 192.168.1.112/29 0.0.0.0                  0         32768 i
*> 192.168.1.120/29 0.0.0.0                  0         32768 i
*> 192.168.1.136/29 192.168.1.102           60         32768 ?
*> 192.168.1.144/29 192.168.1.110           60         32768 ?
*> 192.168.1.152/29 192.168.1.110           60         32768 ?
*> 192.168.1.160/29 192.168.1.110           60         32768 ?
*> 192.168.1.176/29 192.168.1.110           60         32768 ?
*> 192.168.1.184/29 192.168.1.110           60         32768 ?
*> 192.168.1.192/29 192.168.1.110           60         32768 ?
*> 192.168.1.200/29 192.168.1.102           60         32768 ?
R6#
```

We can see that R6's BGP neighbor is being detected!!

```
Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.7.1.101      4   200      27      27       31    0    0 00:18:23       11
R6#
```

It is the same thing over here in R8 for topology A it is able to see every single route from and to every network through redistribution.

```
   Network          Next Hop            Metric LocPrf Weight Path
*> 10.7.1.0/24      20.0.1.101                             0 200 ?
*  20.0.1.0/24      20.0.1.101               0             0 200 ?
*>                  0.0.0.0                  0         32768 ?
*> 92.1.16.0/24     20.0.1.101               0             0 200 i
*> 192.168.0.0/29   192.168.0.206       307200         32768 ?
*> 192.168.0.8/29   192.168.0.198       307200         32768 ?
*> 192.168.0.16/29  192.168.0.198       284160         32768 ?
*> 192.168.0.24/29  192.168.0.190       284160         32768 ?
*> 192.168.0.40/29  0.0.0.0                  0         32768 ?
*> 192.168.0.184/29 0.0.0.0                  0         32768 ?
*> 192.168.0.192/29 0.0.0.0                  0         32768 ?
*> 192.168.0.200/29 0.0.0.0                  0         32768 ?
*> 192.168.1.0/29   20.0.1.101                             0 200 100 ?
*> 192.168.1.8/29   20.0.1.101                             0 200 100 ?
*> 192.168.1.16/29  20.0.1.101                             0 200 100 ?
*> 192.168.1.24/29  20.0.1.101                             0 200 100 ?
*> 192.168.1.32/29  20.0.1.101                             0 200 100 ?
*> 192.168.1.96/29  20.0.1.101                             0 200 100 i
*> 192.168.1.104/29 20.0.1.101                             0 200 100 i
*> 192.168.1.112/29 20.0.1.101                             0 200 100 i
*> 192.168.1.120/29 20.0.1.101                             0 200 100 i
*> 192.168.1.136/29 20.0.1.101                             0 200 100 ?
*> 192.168.1.144/29 20.0.1.101                             0 200 100 ?
*> 192.168.1.152/29 20.0.1.101                             0 200 100 ?
*> 192.168.1.160/29 20.0.1.101                             0 200 100 ?
*> 192.168.1.176/29 20.0.1.101                             0 200 100 ?
*> 192.168.1.184/29 20.0.1.101                             0 200 100 ?
*> 192.168.1.192/29 20.0.1.101                             0 200 100 ?
*> 192.168.1.200/29 20.0.1.101                             0 200 100 ?
```

I had also configured a DHCP server on topology A for easier convenience of IP retrieval for every PC!!

```
R3#sh ip dhcp pool

Pool 2 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 6
 Leased addresses               : 0
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 192.168.0.9          192.168.0.9      - 192.168.0.14      0

Pool 5 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 6
 Leased addresses               : 0
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 192.168.0.17         192.168.0.17     - 192.168.0.22      0
```

We can see that R8's BGP neighbor is being detected!!

```
Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
20.0.1.101      4   200      43      42       29    0    0 00:35:05       20
```

We can see that it is able to ping to the next network through BGP easily!!

```
NAME        : PC2[1]
IP/MASK     : 192.168.1.1/29
GATEWAY     : 192.168.1.4
DNS         :
MAC         : 00:50:79:66:68:01
LPORT       : 20043
RHOST:PORT  : 127.0.0.1:20044
MTU         : 1500

PC2>
PC2>
PC2> ping 192.168.0.1

84 bytes from 192.168.0.1 icmp_seq=1 ttl=56 time=247.535 ms
84 bytes from 192.168.0.1 icmp_seq=2 ttl=56 time=220.626 ms
84 bytes from 192.168.0.1 icmp_seq=3 ttl=56 time=223.836 ms
84 bytes from 192.168.0.1 icmp_seq=4 ttl=56 time=227.393 ms
84 bytes from 192.168.0.1 icmp_seq=5 ttl=56 time=233.317 ms
```

Other configurations remained the same from LAB 6 and nothing changed.!!